Subject: MVP Requirements for the Nutrition Module

1. Core Objective and Inspiration

The primary goal is to develop the Minimum Viable Product (MVP) for the new Nutrition Module. The feature set and user experience for this MVP .We must aim for maximum utilization of its core functionalities and principles
*Red Not part of current scope

2. Feature 1: Comprehensive Food Logging

The module must support robust food logging for all user meals: Breakfast, Lunch, Dinner, and Snacks. This will be facilitated through multiple input methods:

- **Text-Based AI Logging:** Users can type the name of a food (e.g., "Chicken Karahi"). This input will be sent to an AI engine for detection and identification. A confirmation screen will then appear, allowing the user to verify the item, adjust the portion size (increase/decrease), and formally add it to their log.
- **Image Recognition:** Users can take a photo of their meal. The AI must analyze the image to detect the food items present and add them to the log based on this recognition.
- **Barcode Scanning:** A feature to scan the barcode of packaged food items for quick and accurate logging. (Android side?) - Not a blocker
- **Nutritional Label Scanning:** Users must be able to take a picture of the "food items list" (nutritional label) on the back of a product's packaging. The system will parse this image to log the food and its associated data.
- Voice logging (Future) But make it scalable

3. Feature 2: Personalized Daily & Weekly Plans (3 days for MVP)

The system will generate personalized daily and weekly meal plans for the user.

- **Core Synchronization:** These plans must be dynamically synchronized with two key data sets:
    - The user's **Period and Pregnancy** tracking data.
    - All information provided by the user during their **initial onboarding**.
- **AI-Driven Generation:** The plans will be produced by an AI, which must cater to a comprehensive set of user parameters:
    - Country (for local cuisine availability)
    - BMI and BMR
    - Food Preferences
    - Allergies
    - User's specific health goals
    - Target Weight
    - Location
- **Plan Structure:** The output can be a daily/weekly plan or a "phase-wise" plan (e.g., aligned with a pregnancy trimester or menstrual cycle phase). When Plan ends State needs to be shown on main and internal dashboard both and prompt user to create plan
- **Water tracking** to be added and shown explicitly - User can add water input daily and mapping that accordingly with data (Initially can use same structure of water logging in symptoms or by glassed)
- **Weight tracking (manual)** + weekly trend
- **Reminders** to be added (You can add reminder of food and that reminders tell the user of that Simple reminder You can use system reminders for that also like similar concept in periods which we have done)
- **PUSH NOTFICATIONs on device by Android & FCM based for Data Team**
- 2 way syncing (N & Periods , N & pregrancy)
- Suggestion card for meal change & Suggestion cards for period/pregnancy module as if impact on periods/Pregrancy by Nutrition and suggestion cards in Nutrition if period/pregnancy impacting on Nutrition by date , phase ,week , symptoms or others
- Suggestion card to be shown on main and internal dashboards
- Tips to be automated change two way sync of all modules
- On Every meal plan creation Data needs to be passed (Goal based data , Macro and micro nutrients intake data(Which consumed how much and more or less) , Food type taken of what type & frequency and matching with Period phase and day &week if pregnancy and any symptoms mapped also)
- On Meal logging and week passed and month passed analyze and if any impact on period and pregnancy suggestion card in that
- Alert generation syncing periods and pregnancy and diseases it can bring

- Alert of baby and mom
- Alert of female body
- Hormonol balance
- Previous Food logging of current week , day, month & overall, time of food or type of food and mapping with period and pregrancy days ,weeks , phase symptoms & others , Location to be catered by all means for next meal plan creation. What , How much , When , Type , Where user has eaten (Taking Food type like veg, chicken others & Food When-> Time along with Breakfast , lunch , dinner or & Eat Style like Home , Outdoor , Canned ,packed & logging style via text , pic , barcode , label scanning & Macor micor nutrients)
  Nutrition Module & AI learning

4. Feature 3: Data Collection & Progressive Onboarding

Data collection for personalization will be handled in a two-tiered approach:

- **Initial Onboarding:** Essential questions will be asked during the user's main app onboarding.
- **Contextual Onboarding:** If a user skips the initial onboarding, they will be prompted with these essential questions when they first tap on the Nutrition Module.
- **Deep Personalization:** For more detailed data, a prompt will be displayed at the bottom of the module (e.g., "Want to get personalized information? So let's go for a deep test."). This methodology is inspired by the "Live Sumn" application, as previously shared.

5. Feature 4: Symptom-Reactive Plan Adjustments

This is a critical integration point. The nutrition plan must not be static; it must be fully reactive to real-time user inputs, similar to the existing chatbot synchronization.

- **Direct Association:** The daily and weekly plans must be directly associated with the user's logged symptoms.
- **Dynamic Triggers:** The plan must automatically react and adjust based on specific triggers, such as:
  - A newly logged symptom (e.g., "Bloating," "Heavy Spotting," "Cramps").
  - A change in the user's Period or Pregnancy phase/week.
  - Any other critical health alert.
- **Example Workflow:** If a user logs "Cramps," the daily and weekly plan must autonomously adjust its recommendations to include foods and nutrients that help manage cramps. (suggestion card & tips)

6. Implementation of Reactivity (Plan vs. Tip)

When a symptom is logged, the system must react. There are two potential implementations:

1. **Option A (Basic):** Insert a specific "Food Tip" or "Nutrition Tip" into the existing plan (e.g., "You logged cramps. Here is a tip to manage it...").

2. **Option B (Preferred):** Dynamically modify and regenerate the *entire* daily plan based on the newly logged symptom.

Decision: The most preferable outcome is **Option B**, as changing the full daily plan offers a much better and more integrated user experience. If this is not easily possible for the MVP, Option A is the fallback. However, a reaction *must* happen. The preference is to pursue the full plan adjustment.

IMPORTANT

- **Alerts(Specific)**Multiple on maindashbaord and internal specific to that module,
- **Suggestion Cards(synced)** On Main & internal Dashbord (Impact on or by Nutrition),
- **Dynamic Cards (Specific)** Pregrancy specific , Period Specfic , Nutrition specific also synced with each other (Main & Internal Both)
- **Tip**ed with each other but model focuesd
- **Health Checs** Sync**ker** Synced with each other but model focuesd
- **Chatbot** Synced with each other (combined and module trigger
  ->Common for all modules
- Health Alerts to be shown on internal (Nutrition specific) and main dashboard Combine of all Modules (Now in every Module Period , Pregnancy and Nutrition their own health alerts will be shown and on main dashboard combined stacked)
- Health Checker to be shown on internal dashboard Nutrition Specific
- Tips on Main and internal dashboard to changed automatically synced with all modules
- Alerts are separate & Suggestion cards are separate.
- Chatbot also adding Nutrition sync - Chatbot trigger area from nutrition module then catering chatbot with Nutrition perspective - Adding One meal question (MainDashboard combine landing screen if navigated from period or pregnancy their specific but combined context of all modules in app and if navigated from Nutrition module then Nutrition specific but combine context.
- Chatbot trigger on meal logging as well.
  As per previous mapping Nutrition.
- Vitals to be shown and utilized in the app adding watch integration in Nutrition module onboarding and utilization of vitals in Nutrition module (showing vitals in settings or bottom of main dashboard)
- Multiple Alerts can be produced and shown in internal or main dashboard

7. Default State

By default, users will be provisioned with a "standard" daily and weekly plan. This standard plan will then serve as the baseline that is subsequently personalized and adjusted based on the user's unique data (onboarding, symptoms, goals).

Programs -

Add Programs, user can choose the multiple programs from the list.
Plan Switching: Users are not limited to the Standard Plan. There will be a library of **Dietary Plans/Programs** they can choose from, accessible via a "Change Plan" or "Plans" button:
AI to suggest programs as per the user from list like tag of recommended on those.

- When the user taps to explore plans, they navigate to a **Meal Plans Library** page. Here, multiple plan types are listed, each with a brief description. Examples of plan types:
    - **Standard Balanced Plan:** A general healthy eating plan (~maintains weight, balanced macros).
    - **Weight Loss Plan:** A plan with a calorie deficit (e.g. 20% below maintenance) and higher protein to promote fat loss while preserving muscle.
    - **Weight Gain or High-Calorie Plan:** For users looking to gain weight or who have high activity levels, with a calorie surplus and nutrient-dense foods.
    - **PCOS-friendly Plan:** Lower in refined carbs and sugar, higher in fiber and anti-inflammatory foods, to help manage insulin levels and hormones.
    - **Pregnancy Plan:** Ensures adequate calories and emphasizes key pregnancy nutrients (folate, iron, calcium, protein, DHA, etc.), tailored per trimester. (This might even be subdivided: first trimester plan (with ~no extra calories, focuses on folate), second trimester plan (extra ~300 kcal, focus on iron, calcium), third trimester plan (extra ~450 kcal). The app can auto-select the right one based on pregnancy week.)
    - **Postpartum Plan:** (If applicable after pregnancy, could be a plan for recovery and lactation support, high in protein and fluids.)
    - **Pre-conception Plan:** For those trying to conceive, a plan rich in fertility-supporting nutrients.
    - **Gluten-Free Plan, Vegetarian Plan, etc.:** Dietary preference based plans if known (these might be more like filters applied to any plan).
    - **Special Event Plans:** e.g. "Wedding Prep Plan" (often similar to weight loss but maybe with specific timelines), or "Exam Energy Plan" (for students needing brain food during exams, high omega-3 and

antioxidants).

- **Medical Diet Plans:** (if applicable, e.g. for users with diabetes, a low-GI plan; not mentioned explicitly, but PCOS is one example given).
- Grocery list addition

Women-Centric Nutrition Programs in the app but if program selected then producing as programs length

- Sugar Detox Reset (14 Days)
- Low-Sugar Lifestyle Program (21 Days)
- Hormone Balance Nutrition Program
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- Anti-Inflammatory PCOS Meal Plan
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- Postpartum Recovery Nutrition Program
- Breastfeeding Support Nutrition Plan
- Thyroid & Hormone Support Diet Plan
- Weight Management for Women Program
- Menopause Balance Nutrition Program

The Nutrition Module is a core component of the Women's Health app, providing personalized nutrition tracking, guidance, and planning for female users. It integrates tightly with the app's other modules (Period Tracking, Pregnancy, and Fitness) to tailor recommendations and analytics to each user's current life stage and health context. The module empowers users to log their meals and activities, monitor calorie and nutrient intake, receive AI-driven tips and insights, follow meal plans, discover healthy recipes, and track progress over time. The ultimate goal is to help users make informed dietary choices that support their menstrual health, pregnancy, fitness goals, and overall well-being.

Key Features Summary:

- **Context-Aware Personalization:** Nutrition goals and content adapt to the user's status (e.g. menstrual phase, pregnancy week, PCOS or other conditions, fitness activity level).

- **Dashboard with Goals & Summary:** At-a-glance daily calories and macronutrient tracking (with weekly and monthly views) including targets, consumed calories, burned calories, and remaining balance. Dynamic feedback (color-coding, alerts) shows whether the user is on track, over, or under goals.

- **Meal Logging & Tracking:** Easy logging of food intake for Breakfast, Lunch, Dinner, and Snacks. Multiple logging methods are supported – manual entry (with AI assistance), barcode scan, meal photo recognition, and adding from recipe suggestions. Logged items and totals display for each meal.

- **AI Meal Plans:** A personalized "Today's Plan" suggests meals for the day (and up to 7 days ahead) based on user preferences and goals. Users can choose from various meal plan programs (e.g. Standard, Weight Loss, PCOS-friendly, Pregnancy-specific) and see a day-by-day breakdown of planned meals and recipes.

- **Recipe Discovery:** Integrated library of healthy recipes with filters (meal type, calories, macros, micronutrients, etc.) and personalization based on the user's context and preferences. Users can view recipe details (ingredients, nutrition info, cooking steps) and easily add recipe items to their meal log or save favorites.

- **Intelligent Tips & Insights:** Daily AI-driven nutrition tips and educational content cards ("Precise Insights") are presented, tailored to the user's current cycle phase or pregnancy stage, logged symptoms, and goals. A "Health Checker" system analyzes the user's dietary habits (e.g. high sugar, low iron) and conditions (PCOS, etc.) to warn of potential issues or suggest improvements.

- **Progress & Reports:** A calendar and analytics view allows users to review past nutrition data and see trends over days, weeks, and months. Color-coded calendar entries indicate adherence to goals. Users can generate and download detailed nutrition reports (daily, weekly, monthly) – including calorie intake, macro/micronutrient breakdowns, weight changes, and AI-generated insights on health impacts – for personal review or to share with healthcare providers.

- **Weight & Fitness Integration:** The module incorporates weight tracking and fitness data – users can log weight and activities/exercises. Calories burned from workouts (recorded in the Fitness module or via integration) are accounted for, yielding net calorie intake. Weight insights (e.g. weekly weight change) are displayed, and users are prompted to log weight regularly.

- **Third-Party Integrations:** Support integration with external services and devices – e.g. sync with Apple Health/Google Fit for exercise and nutrition data, use third-party food

databases for nutrition info, and leverage AI services for barcode recognition and meal photo analysis.

- **Chatbot Support:** Users can consult a built-in AI chatbot for personalized nutrition assistance – asking questions about their diet, receiving suggestions (e.g. “What snack can I have under 100 calories?”), or clarifying nutrition info in their logs.

(Note: This PRD focuses on the Nutrition module features. Other app modules (Period, Pregnancy, Fitness) are referenced only to explain integration points. Monetization, technical implementation details (database), and user persona profiles are outside the scope of this document.)

Contextual Personalization with Period, Pregnancy & Fitness

The Nutrition module dynamically adapts to each user’s personal context, leveraging data from the Period and Pregnancy modules as well as the Fitness module:

- **Period Module Integration:** If the user is in period-tracking mode (not pregnant), all nutritional guidance is tailored to their menstrual cycle. The app knows the user’s current phase – e.g. menstruation, follicular phase, ovulation (fertile window), luteal phase, or if their period is late. It also considers user-specific conditions or goals:
    - *Cycle Phase Recommendations:* During each phase, nutrition tips and suggested foods are adjusted. (For example, during menstruation the app might encourage iron-rich foods to replenish iron levels, or during the luteal phase suggest magnesium-rich foods to ease PMS symptoms.)
    - *Fertility Goals:* If the user indicated “trying to conceive” or “avoiding pregnancy” in the period module, nutrition content adapts accordingly. E.g., a user trying to conceive might see tips on prenatal vitamins (folic acid) and an optimal diet for fertility.
    - *Health Conditions:* If the user has marked conditions like **PCOS (Polycystic Ovary Syndrome)**, **endometriosis**, thyroid issues, or other relevant health concerns, the nutrition plans and tips account for these. For instance, the app might offer a special PCOS-friendly meal plan (lower in simple carbs and sugar) and show health checker alerts if the user logs foods high in sugar or processed carbs (since these can exacerbate PCOS symptoms).
    - *Menstrual Symptoms Logging:* Symptoms logged in the period module (e.g. cramps, bloating, mood swings) can trigger tailored nutrition tips or article

content. For example, if a user logs migraine during menstruation, the insights card might suggest hydration and foods rich in magnesium or B6 to help. Logged PMS symptoms could lead to recipe suggestions that alleviate those symptoms.

- **Pregnancy Module Integration:** If the user is in pregnancy mode (period tracking is inactive during this time), the Nutrition module adjusts to support a healthy pregnancy:

    - *Pregnancy Week & Trimester:* The app is aware of the current week of pregnancy and trimester. Nutritional targets (calories and key nutrients) and recommendations change as the pregnancy progresses. For example, in the first trimester the focus might be on folate and managing nausea with smaller meals, while in the second trimester calorie needs increase and iron, calcium, and protein become higher priority.

    - *Pregnancy-safe Guidelines:* Meal suggestions and recipes exclude any foods that are unsafe for pregnancy (such as unpasteurized cheeses or certain fishes). The Health Checker will flag if any logged item is not pregnancy-safe, and the app will provide alternatives.

    - *Symptom Integration:* Symptoms from pregnancy tracking (e.g. morning sickness, heartburn, fatigue) can reflect in nutrition tips. For instance, if a user reports heartburn, the daily tips might advise avoiding spicy or acidic foods that day, or suggest ginger if experiencing nausea.

    - *Condition Considerations:* If the user has pregnancy-related conditions like gestational diabetes or hypertension, the module will adjust recommendations (e.g. lower sugar/carb plans for gestational diabetes, lower sodium tips for hypertension) and provide appropriate alerts.

- **Fitness Module Integration:** The Nutrition module works hand-in-hand with fitness tracking to give a full picture of energy balance and to encourage a holistic healthy lifestyle:

    - *Calories Burned:* Any exercise logged in the Fitness module (or via connected wearable devices) contributes to the “burned calories” metric on the nutrition dashboard. The daily calorie card shows **Net Calories = Calories Consumed – Calories Burned**, helping the user understand their caloric balance. For example, if the daily goal is 2000 kcal and the user ate 2100 kcal but burned 300 kcal through exercise, the dashboard will show they are still under net goal by 200 kcal.

    - *Exercise Mapping:* The type and intensity of exercises can influence nutritional advice. If a user did an intense workout, the app might suggest a higher protein intake in tips or recommend a recovery smoothie recipe. The insights content

might also highlight nutrition topics relevant to fitness (e.g. protein for muscle repair, hydration after workouts).

- *Activity Logging from Nutrition:* The Nutrition interface will provide a shortcut to log an activity without needing to switch to the Fitness module. For example, on the nutrition dashboard or via the quick "+" add button, the user can add a workout (selecting from common exercises or syncing from a device). This convenience ensures their calorie burn is up-to-date when planning meals.

- *Fitness Goals Alignment:* If the user has set a fitness goal (e.g. training for a 5K, or a weight loss goal through exercise), the meal plans and tips will reflect that. For instance, a user focusing on muscle gain might be offered a high-protein meal plan, whereas a user with a weight loss goal will see a calorie-deficit meal plan with appropriate nutrient balance.

- *Cross-Module Prompts:* The app will occasionally prompt the user with cross-module suggestions. For example, if the user's calorie intake is high on a given day, a prompt might encourage doing a bit of exercise (linking to a quick workout in the Fitness module) to balance out. Conversely, after a big workout, it may remind to increase hydration or certain nutrients.

Mutual Exclusivity: A user can either be in Period mode or Pregnancy mode at a given time (never both). The Nutrition module detects this and applies the relevant context. If a user transitions (e.g. becomes pregnant, or postpartum returns to period tracking), the module will switch its context appropriately. This context is prominently indicated in the Nutrition UI (for example, showing "Pregnancy Mode – Week 20" or "Cycle Day 15 – Ovulation Phase" at the top of the nutrition dashboard or in the plan section), so the user is aware that their data and recommendations are tailored for that situation.

Goals: Add Goals in nutrition module and also allow logging of vitamins & supplements

User Interface & Dashboard Overview

Upon entering the Nutrition module, the user is greeted with a dashboard summarizing their key nutrition info and providing entry points to all major features. The design emphasizes clarity, quick status checks, and easy logging. The dashboard consists of several sections (cards) stacked vertically, which the user can scroll through:

1. Calorie Goal Card (Daily/Weekly/Monthly Overview)

At the top of the Nutrition dashboard, a prominent **Calories Summary Card** displays the user's calorie intake status against their goals:

- **Toggle Views:** By default, it shows the **Daily** goal progress, but the user can switch the view to **Weekly** or **Monthly** aggregate progress. (Tapping on the card or a toggle control cycles through daily, weekly, monthly summaries.)

- **Daily View:** Shows today's calorie target and the breakdown:

    - **Target Calories** – the daily calorie goal (personalized based on user's profile and goals, and adjusted for pregnancy if applicable).

    - **Consumed (Eaten) Calories** – total calories from all foods logged so far today.

    - **Burned Calories** – total calories burned through exercise/activity today (sourced from the fitness module or manual entry).

    - **Remaining Calories** – the difference: `Target − (Consumed − Burned)` i.e., how many calories left to eat (or if negative, how many over the goal).

- **Weekly/Monthly View:** When toggled, the card aggregates similar info for the current week or month:

    - **Target** – the cumulative target (e.g. weekly target = daily target * 7).

    - **Consumed** – total calories eaten in that period.

    - **Burned** – total calories burned in that period.

    - **Net / Remaining** – net calories vs target.

    - It may also show an average per day or how many days met goal out of the total.

- **Visual Representation:** The card uses visual cues:

    - A ring or bar indicator that fills based on percentage of goal met.

    - Color coding: **Green** when the net intake is within the acceptable range of the goal, **Red** if significantly over the goal, **Orange/Yellow** if under the minimum threshold. (Exact thresholds can be defined by nutritional guidelines or user settings; e.g., >10% over goal triggers red, >10% under triggers orange.)

    - For weekly/monthly, a small graph or series of bars might show each day's status in brief, or just color coding for overall performance.

- **Dynamic State Messages to be shown on internal and Main dashboard both:** The calorie card also provides contextual messages and states depending on user's logging:

    - **No Data Logged (Empty State):** If the user has not logged any food for the day yet, the card will prompt: "No meals logged yet today." It may encourage the user to start logging (e.g. "Let's add your breakfast to begin tracking calories."). For weekly/monthly, if some days are unlogged, it might say "Incomplete data for X days."

    - **Under Target:** If by end of day (or at the current time) the consumed calories are well below target, a message appears like "You're 400 kcal below your goal. Try to eat a bit more to meet your nutritional needs." This gentle nudge encourages adequate intake, important for women's health (especially if pregnant or managing cycle-related energy needs).

    - **Over Target:** If consumed calories exceed the goal, a message might say "You've exceeded your daily goal by 250 kcal." along with a tip such as "Consider a light exercise or a lighter dinner to balance out." (This ties in the fitness integration by suggesting action when over target.)

    - **On Track:** If consumption is on track, the card can show a motivational message: "Great job – you're on track with your calories today!" possibly with a green check or smile icon.

    - **Missing Log Days:** If the user has not logged anything for previous days, a subtle alert is shown: e.g. "No data for the last 2 days." This text would appear beneath the daily summary. It serves as a reminder, and if tapped, it navigates to the Calendar view where the user can back-fill those days' meals. (In weekly view, if some days are missing data, a similar note appears.)

- **Alerts & Notifications:** The card will also highlight if there are specific alerts. For example, if the user consistently undereats or overeats for several days, a small warning icon might appear with a note like "Consistently under target this week" – encouraging the user to review their diet or seek advice. These alerts consider the user's context (for instance, a pregnant user under-eating will see a more urgent reminder about meeting nutritional needs, while a non-pregnant user trying to lose weight might see a gentler note about safe calorie deficits).

A **"See All" or "Details"** button/link on this card will lead to a more detailed analysis page (covered later in **Progress & Analytics**), where the user can further explore their calorie intake history, see charts and download reports.

2. Macronutrients Card (Carbs, Proteins, Fats)

Directly below the calorie summary, the **Macros Card** provides a snapshot of the user's macronutrient intake distribution and goals:

- **Macronutrient Targets:** The app calculates recommended daily targets for the three macronutrients – **Carbohydrates, Proteins, Fats** – based on the user's profile (could follow a default ratio like 50% carbs, 20% protein, 30% fats for a maintenance diet, adjusted if the user has chosen a specific diet plan or context like higher protein for fitness, etc.). These targets are displayed as grams or as percentage of total calories.

- **Consumed Amounts:** The card shows how much of each macro the user has consumed so far today (in grams and/or % of the target). For example: "Carbs: 150g of 250g target (60%)".

- **Visual Graph:** A bar graph or pie chart could illustrate the macro breakdown:

    - If using a pie, it shows the proportion of calories from carbs/protein/fat relative to each other and highlights whether the balance meets the recommended ratio.

    - If using bars, each macro has a bar indicating progress towards its goal (with the bar turning green/yellow/red similar to calories).

- **State Feedback:** Similar to the calorie card, the macros card gives feedback:

    - If one macro is far below target (e.g. protein intake very low by end of day), a note might say "Low protein intake – consider a protein-rich snack."

    - If one macro is excessive (e.g. fats way above target), "High fat intake – try leaner food choices for the next meal."

    - If everything is within range, "Macronutrient balance looks good today."

- **Missing Data:** If no foods are logged yet, the card remains mostly blank or shows "No nutrition data yet."

- **Daily vs Extended:** By default, it's daily. The "See All" from this card also goes to the detailed Nutrition Analytics page. On that page, users can toggle daily/weekly/monthly views for macros as well (e.g. average macro distribution over a week).

- **Integration with Context:** The macros recommendation might adjust with context. For example, during pregnancy, protein and certain healthy fats goals might be higher. If the user is on a low-carb plan or has diabetes, the carb target might be custom. The card reflects those personalized targets.

- **Micronutrient Teaser:** The macronutrient card might also include a small teaser or summary about micronutrient intake (e.g. “Fiber: 18g of 25g” or “Sodium: 2300 mg (high)” for the day) to hint at detailed info available elsewhere. However, full micronutrient details would be in the expanded analytics or in each food/recipe detail.

Women-Centric Nutrition Programs in the app but if program selected then producing as programs length

- Sugar Detox Reset (14 Days)
- Low-Sugar Lifestyle Program (21 Days)
- Hormone Balance Nutrition Program
- PCOS-Friendly Diet Program
- Anti-Inflammatory PCOS Meal Plan
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- Gestational Diabetes Meal Program
- Postpartum Recovery Nutrition Program
- Breastfeeding Support Nutrition Plan
- Thyroid & Hormone Support Diet Plan
- Weight Management for Women Program
- Menopause Balance Nutrition Program

Note: Stand Alone one module can be selected and upselling of other module on main and internal dashboard. Detailed Nutrition onboarding

If only Nutrition Module selected all nutrition module questions Default & progressive all and first time user experience handling of Nutrition User as well -> Guide Onboarding Tutorial as well
Upselling Logic & Missed usage logic:
If Period selected then upselling of Nutrition on Main and Internal dashboard & if Nutrition Module selected upselling of Period and pregnancy on main and internal dashboard

And If both selected then upselling that module which is not being used.
Nutrition active but period no logging

Period/Pregrancy Active but nutrition not logging

Last 3 days No nutrition Logging - Upsell Nutrition
Last 5 Days No Period/Pregrancy logging - Upsell period or Pregrancy

Rule A — If Period selected (and Nutrition not selected)

Show **Nutrition upsell** on:

- **Main dashboard**
- **Period internal dashboard**
  Only if:
- Period is **Activated**, AND
- user has had **≥2 sessions** OR **≥1 log** (avoid showing instantly on day 0)

Primary CTA: “Add Nutrition” → Nutrition onboarding

Rule B — If Nutrition selected (and Period/Pregnancy not selected)

Show **Cycle Tracking upsell (Period or Pregnancy)** on:

- **Main dashboard**
- **Nutrition internal dashboard**
  Only if:
- Nutrition is **Activated**, AND
- user has completed at least **one** nutrition action (goal set / first meal log / plan viewed)

Primary CTA: “Track your cycle” → Choose: Period / Pregnancy

Rule C — If Both selected

Upsell **the module that is NOT being used** (your exact requirement).

C1: Nutrition selected & Period selected

If:

- Nutrition is **LoggingActive = true**
- Period is **LoggingActive = false**
  → Upsell **Period logging habit** (NOT module selection)

Surfaces:

- Main dashboard
- Nutrition internal
- Period internal (soft nudge inside)

CTA examples:

- “Log today’s symptoms”
- “Quick period check-in (10 sec)”

C2: Period/Pregnancy selected & Nutrition selected

If:

- Period/Pregnancy is **LoggingActive = true**
- Nutrition is **LoggingActive = false**
  → Upsell **Nutrition logging habit**

Surfaces:

- Main dashboard
- Period/Pregnancy internal
- Nutrition internal (if they open it but don’t log)

CTA examples:

- “Log your breakfast”
- “Start a 3-day meal streak”

Nutrition v2

Onboarding and improvements

Overview

This section documents iterative improvements to the Nutrition onboarding flow and module entry logic following the implementation of Version 1. These changes aim to:

- Reduce user confusion during onboarding
- Ensure critical personalization inputs are captured
- Improve safety around weight goal selection
- Strengthen data integrity for meal plan generation

All updates described below apply to the Nutrition onboarding flow unless otherwise specified.

1. Single Module Selection in Onboarding

As of right now users will select only one primary module during onboarding.

The selected module determines the onboarding flow shown to the user.

Rationale

- Reduces cognitive overload during onboarding
- Ensures focused personalization

- Prevents cross-module branching confusion
- Improves onboarding completion rate

2. Remove “Period or Pregnant” Screen from Nutrition Onboarding

Current State

Nutrition onboarding includes a screen asking users to choose “Period” or “Pregnant.”

Update (V2)

The “Period or Pregnant” screen will be removed from the Nutrition onboarding flow.

Clarification

Pregnancy or cycle status will be determined from:

- Module selection (if user selects Pregnancy module)
- Or existing app-level user status

Nutrition onboarding should not independently determine reproductive state.

3. Updated Nutrition Onboarding Flow Logic

3.1 Target Weight Screen Conditional Logic

The target weight screen will only appear if the selected wellness goal is:

- Losing weight
- Gaining weight

It will NOT appear if the goal is:

- Maintain weight
- General wellness

- Any non-weight-based goal

When setting target weight, the user can select a goal that is 30% more or less than their current weight to prevent any extreme goal setting and timelines.

3.2 Weight Loss Slider Configuration

For weight loss:

- Slider range: **0.1 kg/week to 1.0 kg/week**
- Default value: **0.5 kg/week**
- 0.1–0.5 kg/week = Slow to moderate
- 0.6–1.0 kg/week = Moderate to fast

Visual Feedback Logic

- 0.1–0.7 kg/week → Display text in neutral/safe colour
- 0.8 kg/week and above →
    - Display text in orange
    - Show supporting text:

  “Faster weight loss may not be sustainable for everyone.”

3.3 Weight Gain Slider Configuration

For weight gain:

- Slider range: **0.1 kg/week to 1.0 kg/week**
- Default value: **0.5 kg/week**
- 0.1–0.5 kg/week = Slow to moderate
- 0.6–1.0 kg/week = Moderate to fast

Visual Feedback Logic

- 0.1–0.7 kg/week → Neutral/safe colour
- 0.8 kg/week and above →
    - Display text in orange
    - Show supporting text:

> “Rapid weight gain may lead to increased fat storage rather than muscle.”

Rationale for Weight Slider Adjustments

- Encourages safer, sustainable rate selection
- Adds behavioural nudging without blocking choice
- Improves trust by signaling caution at extreme values

4. Skip Functionality in Nutrition Onboarding

Update (V2)

All onboarding inputs are mandatory except:

- Specific diet preference selection (this remains skippable)

Mandatory fields:

- Age
- Weight
- Height
- Wellness goal
- Target weight (if applicable)

Rationale

- Ensures accurate calorie calculations
- Ensures meal plan accuracy
- Prevents incomplete personalization
- Reduces future data correction friction

5. Default Meal Plan Generation

According to the recommendations made by the AI team, target calories will now be calculated based on a default meal plan, which will be generated on the backend once the onboarding data has been collected and will be personalised to that information. This will happen while the user is shown the generating plan screen. Once generated it will be used to calculate the target calories, which will be shown to the user on the dashboard as soon as they land on it.
The user will also see the default meal plan on the nutrition dashboard, which they can follow. Once the plan expires the user can generate a new one by answering the progressive onboarding questions, whose input will also be taken into account while generating the new meal plan.

6. Addition of Food Allergies Screen

Update (V2)

A new Food Allergies screen will be added to Nutrition onboarding, which was previously only added to the progressive onboarding when the user clicks on meal plan generation.

This input is required to ensure meal plan safety and personalization accuracy.

Placement in Onboarding Flow

Age, wellness goal, specific diet preference (skip option only on this screen), allergies, height, weight, target weight (only for those who select gain or lose weight), speed of gaining or losing weight (only for those who select gain or lose weight), generate plan.

If a user has already answered the allergy question during the onboarding, they should not be shown the question in the progressive onboarding again when they generate a new meal plan. If they skipped it in the onboarding, the question should be mandatory during the progressive onboarding.

6. Remove “None” Option from Wellness Goal Screen

Update (V2)

The “None” option will be removed from the Wellness Goal selection screen.

Users must explicitly select a goal.

Food Logging Edge Cases

1. Multi-Item Image Detection Handling

Requirement

If AI detects multiple food items in a single image:

- Each detected item must be presented as an individually editable entry.
- Entries should be shown in a structured list (individual screens or stacked editable cards — design-specific layout flexible).
- The system must not assume a single combined serving unless AI explicitly returns one item.
- AI will detect all diverse items individually. For text based items, AI will infer the serving based on the text, and for image based items, the serving will be shown as 1 by default, which the user can then adjust.

Goal

Ensure users can:

- Review each detected item independently.
- Correct mistakes before logging.
- Maintain trust even when AI partially misidentifies items.

2. Editable Entry State (Per Food Item)

Each detected item must allow the following actions before confirmation:

2.1 Edit Food Name

- User can edit the detected food name.
- Upon editing:
    - User must have the ability to recalculate calories/macros based on the updated name.
    - This may trigger a regeneration API call.

2.2 Edit Calories Manually

- User can manually adjust calorie value.
- Manual edit should override AI estimate unless recalculated.

2.3 Adjust Serving Size

- Serving increment must support 0.5 increments.
- Example: 0.5 → 1.0 → 1.5 → 2.0.

2.4 Regenerate / Recalculate Option

Calories will need to be updated in the following cases:
- If the user updates the serving size
- If the user edits the name

In case of the user updating the serving size, we don't need to call the API again and incur cost. The current calories should be stored in the backend or cached and when the user increases the serving, backend should use the same count of calories and add them based on the increase in the serving size.

In case the user edits the name, then the API needs to be called again (this is on the edit screen before adding the item). Since calories don't usually need to be updated over minor name edits and only need to be updated in case AI has misidentified the item, the user will be shown a message upon editing the name, saying ''You have changed the name of the item. Would you

like femverse AI to recalculate the calories based on the new name?'' If they say yes only then the calories need to be recalculated.

Note: We will track the usage of the recalculate feature in the first build. If we later see that very few recalculations are being done, we will remove the button and enable automatic API call upon name edits.

3. Entry Management

Within the same confirmation flow, user must be able to:

- Delete an entry
- Add a new food item manually
    - New item becomes part of the same logging batch
    - Same editable rules apply

4. AI Transparency for Incorrect Detection

If user selects "Flag incorrect calories" (or similar):

- Show what AI originally detected from the image.
- Allow user to see:
    - The detected food name(s)
    - Any detection context returned by AI

Purpose:

- Reduce unnecessary flagging.
- Increase transparency.
- Build user trust.

5. Trust & Expectation Labels

On the confirmation screen (image logging only), display:

> “Estimated based on typical serving. You can adjust.”

Additionally, include image guidance messaging:

> “For best results, take clear images in good lighting.”

These labels reinforce:

- AI estimate nature
- User control
- Accuracy expectations

6. Error Handling

If AI fails during image processing:

- Show clear reason for failure.
- Provide:
    - Retry option
    - Option to log via text instead

No generic error states.

7. State Preservation (Caching)

- Main food logging screens should preserve previously loaded states.

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1. Tap-and-Hold for Fast Increment

Update

Users can tap and hold the "+" or "−" buttons to continuously increment or decrement serving size.

Behavior

- Continuous increment begins after a short press delay (~300ms).
- Increment increases progressively (optional acceleration after 1–2 seconds).
- Releasing the press immediately stops the change.

Applies To

- All serving size adjustment controls across:
    - Image-detected foods
    - Text-entered foods
    - Barcode-scanned foods

Edge Cases

- How will calorie recalculation work here?

2. Manual Numeric Entry (Hidden Behind Edit Icon)

Update

Users can manually enter a numeric quantity value by tapping an edit icon next to the serving size field.

Behavior

- Opens numeric keypad.
- User can input decimal values.

- Validation occurs on submit.

Validation Rules

- Must be greater than minimum allowed value.
- If invalid:
    - Show inline error message.
    - Do not save value.

After Confirmation

- Quantity updates immediately.

Period Onboarding Calendar – Month Display Logic (V2 Update)

Calendar UI Update

During period data onboarding:

- Replace the scrollable multi-month calendar design.
- Display only a single month view by default as shown in figma.
- Users can manually navigate to previous months if needed.
- Forward navigation (future months) is not required in onboarding.

Default Month Display Logic

The month shown on entry will be determined by today's date:

- If today's date is 14th or earlier:
    - Show the previous month by default.

- If today's date is 15th or later:
    - Show the current month by default.

Rationale for Logic

- If the date is early in the month (≤14), the user's last period is more likely to have started in the previous month.
- If the date is later in the month (>14), the user's last period is more likely to have started in the current month.
- Most users will not need to navigate backward.
- Only irregular-cycle users may need to manually navigate to older months.

Settings - updating user information and nutrition goals